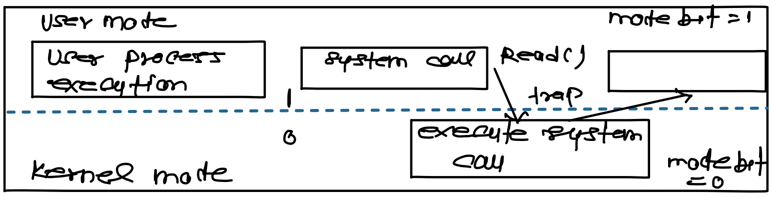
Operating System:-

System call:- A system call is a way for program to interact with the opearenting system.

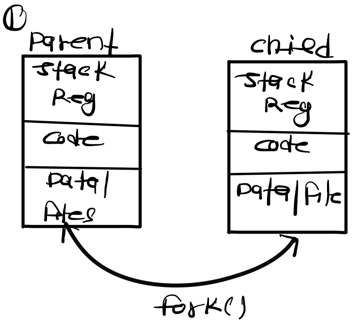
Dual mode operation:-

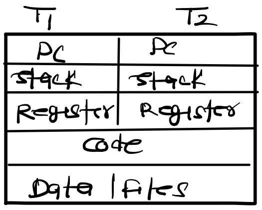
User vs Kernel mode:-



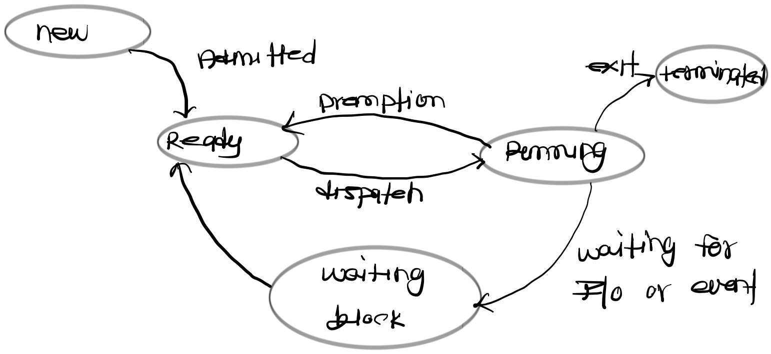
→ used for system protection.

Process :-	threads: -
①	①
② OS treats different process differently and Different process have different copies of	② A thread is a Light weight process and each thread comprises its own stack, program counter, register & only the ~~Addres~~ space of the thread is shared with other.
③ System call involved in process.	③ there is no system call involved.
④ Blocking a process will not block another process	④ Blocking a process will block entire process.

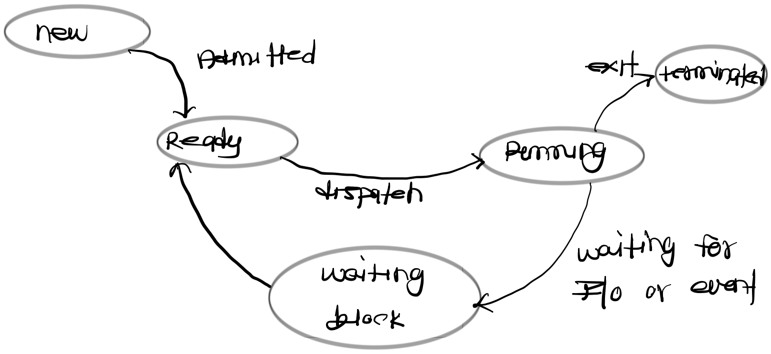




Process states:- (Premptive)



Process states:- (Non- Premptive)



→ Contex switch is done by dispatch

CPU scheduling:-

① FCFS (first come first serve):-

Process	AT	BT	CT	TAT	WT
P_1	0	27	27	27	0
P_2	0	6	33	33	27
P_3	0	6	39	39	33

$\boxed{\text{Avg TAT} = CT - AT}$ $\boxed{WT = TAT - BT}$

Convey effect:- If a heavy (big) process is scheduled first then it delays All other process execution significantly, Hence performance of the system slow down

② SJF (shortest Job First):-

Shedueling criteria:- shedule process with small Burst time
⇓
tie breaker (FCFS)

Process	AT	BT	CT	TAT	WT